Yanzhen Chen

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Education

University of Toronto September 2023 – Expected April 2028

Honours Bachelor of Science in Computer Science (Co-op) | CS GPA: 3.82 *Toronto, ON*

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Databases, Deep Learning

Teaching Assistant: *CSC165*: Mathematical Expression and Reasoning for Computer Science (Winter 2026)

Experience

Bree (YC S21) Toronto, ON

Software Engineer Intern *May 2026 – Present*

- Hardened **Interac e-Transfer** disbursement against duplicate and dropped payments at a fintech serving **800,000+ users**: persisted provider acknowledgements, keyed retries for idempotency, blocked retries on already-acknowledged transfers, and required exact matching on orphaned fees.
- Built webhook receivers and ingestion modules for third-party integrations in **TypeScript** on **AWS Lambda**, propagating retryable failures instead of acknowledging them and guarding against out-of-order delivery and read-replica lag; authored **11 production PostgreSQL migrations**.
- Shipped backend product logic across cloud function controllers and a shared data-access layer, including card-rail routing that redirects unsupported cards to e-transfer, a 24-hour application cooldown enforced at submission, an OCR data model with persistence, and an extended collections pause window.
- Led the migration of Bree's production **React Native** app from React Navigation to **Expo Router** across **78 screens and 230+ call sites**, designing a backward-compatible shim with typed routes, deep-link ownership, wire-encoded params that keep PII out of router URLs, and CI running the **745-test suite** against both navigation backends.

University of Toronto, MarkUs Project Toronto, ON

Open Source Contributor *May 2026 – Present*

- Landed **20 merged pull requests** to MarkUs, an open-source **Ruby on Rails** assignment-management platform used by **15,000+ students** across the University of Toronto, University of Waterloo, and École Centrale de Nantes, improving authentication, instructor workflows, course administration, and testing infrastructure.

IQBank Toronto, ON

Technical Lead & Software Engineer *January 2026 – Present*

- Leading a team of **6 developers**, driving architecture decisions, reviewing pull requests, mentoring contributors, and coordinating frontend and backend development.
- Built full-stack features with **React, TypeScript, Node.js, and PostgreSQL**, including role-based access control, persistent drafts, user profiles, and bulk account creation for an exam-preparation platform serving **300+ students per semester**.

Projects

NovaOS | *TypeScript, React, Vite, Vitest, Turborepo* | **Demo** June 2026 – Present

- Built a browser-based operating systems laboratory as a **16-package TypeScript monorepo**, including a C-like compiler performing parsing, semantic analysis, optimization, and source-mapped code generation.
- Designed a custom **32-bit virtual machine and microkernel** with **37 instructions**, process scheduling, system calls, memory management, and inter-process communication.
- Built a **time-travel debugger** supporting deterministic replay, source-level stepping, **5 breakpoint types**, call-stack reconstruction, and live register and memory inspection.

TorontoGuessr | *TypeScript, Next.js, Tailwind, Supabase, Google Maps API* | **Demo** April 2025 – Present

- Built and deployed a full-stack geolocation game with **Next.js**, **PostgreSQL**, and **Google Street View**, serving **300+ monthly players** with more than **4,000 games played**.
- Designed APIs for randomized Street View generation, persistent scoring, and leaderboards backed by more than **2,000 curated Toronto locations**.

Leadership Experience

President May 2026 – Present

Computer Science Student Union *Toronto, ON*

- Leading a **10-person executive team** serving **2,300+ computer science students**; previously served two years as VP External Relations, securing **$14,300 in sponsorships** and industry partnerships.

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C++, C, Ruby, SQL
Frameworks: React, React Native, Next.js, Node.js, Express, Expo, Ruby on Rails
Infrastructure & Tools: AWS, Docker, PostgreSQL, Supabase, Prisma, Linux/Unix, Git, GitHub Actions, Claude Code